

Lab 3 – Architectural Justification

Project: Freelance Content Creator Escrow Platform **Selected Architecture:** Layered Architecture

Architecture Selection

We chose **Layered Architecture for the Freelance Content Creator Escrow Platform**. The system is organized into Presentation, Business, and Data layers. The Presentation Layer contains the User Interface Component. The Business Layer contains the Order Manager, Deliverable & Review, and Payment Service components. The Data Layer contains the Database Component, while the Payment Gateway is treated as an external system.

Reason 1 – Separation of Responsibilities

Layered Architecture separates user interaction from business processing and data storage. The User Interface handles requests from the Client Sponsor and Content Creator, while the Order Manager, Deliverable & Review, and Payment Service components handle project, deliverable, review, escrow, and payment operations. This separation makes the system easier to maintain and modify without changing unrelated layers.

Reason 2 – Suitable for the Escrow Workflow

The platform follows a structured workflow: a project is created and funded, deliverables are submitted and reviewed, revisions may be requested, and payment is released after approval. Layered Architecture provides clear interfaces between these stages and fits a centralized web-based workflow without requiring the operational complexity of independently deployed services.

Security Advantage

Payment processing is isolated in the Payment Service Component. The User Interface does not communicate directly with the external Payment Gateway. Payment requests pass through the business layer, where validation and authorization rules can be applied before a payment transaction is requested.

Performance Benefit

The architecture avoids unnecessary distributed communication between multiple independently deployed services. Related business operations can communicate through defined interfaces within the application, reducing communication overhead and supporting responsive project, deliverable, and payment operations.

Main Components	Primary Responsibility
User Interface	Client/Creator interaction
Order Manager	Project and workflow management
Deliverable & Review	Submission, review and revision
Payment Service	Escrow and payment processing
Database	Persistent platform and transaction data
Payment Gateway	External payment transaction service